EX:1 Estimation of Characteristic Impedance (Zo), and Propagation Constant (y) of aTransmission Line from Primary Constants

Test Case 1: Estimation of Characteristic Impedance (Zo), and Propagation Constant (y) of aTransmission Line from Primary Constants a) b) c) d) e) Investigating the significance of Characteristic Impedance (Zo), and Propagation Constant (7) of aTransmission Line using the following parameters Resistance =10 (YKm Capacitance = 8 gF/Km Inductance =3.5 mH/Km Conductance =4 gmho/Km Frequency= 2 Khz

CODE:

```
clear;
clc;

R = 10;
L = 0.003;
f = 2 * 10^3;
G = 4 * (10^-6);
C = 16 * (10^-6);

w = 2 * %pi * f;

Z = R + (%i * w * L);
Y = G + (%i * w * C);

Zo = sqrt(Z / Y);

Cl = round(real(Zo));
D = round(imag(Zo));

printf("- Zo = %f + j(%f) ohms\n", Cl, D);

P = sqrt(Z * Y);

a = real(P);
al = round(a * 10000) / 10000;

printf("- Attenuation constant a = %f neper/km\n", al);

b = imag(P);
bl = round(b * 10000) / 10000;

printf("- Phase constant b = %f radians/km\n", bl);
```

OUTPUT:

```
- Zo = 14.000000 + j(-2.000000) ohms
- Attenuation constant a = 0.362100 neper/km
- Phase constant b = 2.776900 radians/km

-->
```

Test Case 2: Examine the impact of resistance and inductance on the significance of Characteristic Impedance (Zo) and Propagation Constant (7) of a Transmission Line at a frequency f=2.5 Khz. Vary the resistance value from 10 0 to 50 0 at the interval of 10 0 with L,G and C values constant and

calculate the Characteristic Impedance and Propagation Constant Vary the inductance values from 3.5 mH to 11.5 mH at the interval of 2.5 mH with R,G and C values constant

CODE:

```
clear;
clc;

R = 10;
L = 0.003;
f = 2 * 10^3;
G = 4 * (10^-6);
C = 16 * (10^-6);

w = 2 * %pi * f;

Z = R + (%i * w * L);
Y = G + (%i * w * C);

Zo = sqrt(Z / Y);

C = round(real(Zo));
D = round(imag(Zo));

printf('- Zo = %f + j(%f) ohms\n', C, D);

P = sqrt(Z * Y);

a = real(P);
al = round(a * 10000) / 10000;

printf('- Attenuation constant a = %f neper/km\n', al);

b = imag(P);

bl = round(b * 10000) / 10000;

printf('- Phase constant b = %f radians/km\n', bl);

```

OUTPUT:

```
Scilab 2025.1.0 Console
- Zo = 14.000000 + j(-2.000000) ohms
- Attenuation constant a = 0.362100 neper/km
- Phase constant b = 2.776900 radians/km

-->
```

Test Case 3: Explore the impact of frequency and capacitance on the significance of Characteristic Impedance (Zo) and Propagation Constant (y) of a Transmission Line. Vary the frequency from 2KHz to 1() KHz at the interval of 2 KHz with L,R ,G and C values constant and calculate the Characteristic Impedance and Propagation Constant . Vary the capacitance

values from 8 pF to 16 pF at the interval of 2 pF with R,G and L values constant

CODE:

```
clear;
clc;

R=10;
L=0.003;
f=2*10^3;
G=4*(10^-6);
C=16*(10^-6);

w=2*%pi*f;

Z=R+(%i*w*L);
Y=G+(%i*w*C);

Zo=sqrt(Z/Y);

C=round(real(Zo));
D=round(imag(Zo));

printf('-Zo = %f + j(%f)ohms\n',C,D);

P=sqrt(Z*Y);

a=real(P);
al=round(a*10000)/10000;

printf('-Attenuation constant a = %f neper/km\n',al);

b=imag(P);
bl=round(b*10000)/10000;

printf('-Phase constant b = %f radians/km',bl);
```

OUTPUT:

```
-Zo = 14.000000 + j(-2.000000)ohms
-Attenuation constant a = 0.362100 neper/km
-Phase constant b = 2.776900 radians/km
-->
```

EX:2

Estimation of Primary Constants (R,L,G &C) from Secondary Constants (Zo &)of an open wiretransmission line

Test Case 1: Estimation of Primary Constants (R,L,G &C) from Secondary Constants (Zo &)of an open wiretransmission line Find the primary constants R,L,G and C for the varying values of Characteristic Impedance (Zo) of a Transmission Line at a frequency f=2 Khz. Vary the Characteristic Impedance valup from 1()() O and take five readings at the interval of 2() O. Keep the Propagation Constant (7) constant

CODE:

```
clear;
clc;

Zo=100.6;
f=2000; //value of Zo as taken in solution
P=0.054*exp(%i*(%pi/(180/87.9)));

w=2*%pi*f;

Z=Zo*P;

R=real(Z);

printf('-Resistance R = %f ohms/km\n',R);

L=(imag(Z))/w;

printf('-Inductance L = %f mH/km\n',L*(10^3));

Y=P/Zo;

G=real(Y);

printf('-Conductance G = %f micromhos/km\n',G*(10^6));

C=((imag(Y))/w)*(10^6);
c=round(C*10000)/10000;

printf('-Capacitance C = %f microfarads/km\n',c);

```

OUTPUT:

```
-Resistance R = 0.199063 ohms/km
-Inductance L = 0.432006 mH/km
-Conductance G = 19.669585 micromhos/km
-Capacitance C = 0.042700 microfarads/km

-->
```

Test Case 2: Find the primary constants R,L,G and C for the varying values of Propagation constant of a Transmission Line at a frequency f=2 Khz. Vary the

Propagation constant value from 200 and talw five readings at the interval of 50 0. Keep the characteristic impedance Zo=100Q constant

CODE:

```
clear;
clc;

Zo=100.6;
f=2000; //value of Zo as taken in solution
P=0.054*exp(%i*(%pi/(180/87.9)));

w=2*%pi*f;

Z=Zo*P;

R=real(Z);

printf('-Resistance R = %f ohms/km\n',R);

L=(imag(Z))/w;

printf('-Inductance L = %f mH/km\n',L*(10^3));

Y=P/Zo;

G=real(Y);

printf('-Conductance G = %f micromhos/km\n',G*(10^6));

C=((imag(Y))/w)*(10^6);
c=round(C*10000)/10000;

printf('-Capacitance C = %f microfarads/km\n',c);

```

OUTPUT:

```
-Resistance R = 0.199063 ohms/km
-Inductance L = 0.432006 mH/km
-Conductance G = 19.669585 micromhos/km
-Capacitance C = 0.042700 microfarads/km

-->
```

EX:3

Evaluation of current in short circuited Lossless transmission line at the receivin end

Test Case 1: Evaluation of current in short circuited Lossless transmission line at the receivin end Find the current at the receiving end for the varying values of the attenuation constant a from keeping the characteristic impedance Zo and the phase constant constant

CODE:

```
clear;
clc;

Zo=600;
a=0.1;
b=0.05;
x=10;
Is=20*(10^-3);

Vr=0;

printf("-Receiving end voltage Vr=0 because the receiving end has been short circuited\n");

P=a+(%i*b);

Ir=Is/(cosh(10*P));

A=real(Ir);
B=imag(Ir);

printf("-Received current is Ir = %f /_%f mA\n",round(abs(Ir)*(10^3)*100)/100,fix(atan(B,A)*180*10/%pi)/10);

```

OUTPUT:

```
-Receiving end voltage Vr=0 because the receiving end has been short circuited
-Received current is Ir = 13.640000 /_-22.500000 mA
```

Test Case 2: Find the current at the receiving end for the varying values of the phase constant attenuation constant a from keeping the characteristic impedance Zo and the attenuation constant a constant

CODE:

```
clear;
clc;

Zo=600;
a=0.1;
b=0.05;
x=10;
Is=20*(10^-3);

Vr=0;

printf("-Receiving end voltage Vr=0 because the receiving end has been short circuited\n");

P=a+(%i*b);

Ir=Is/(cosh(10*P));

A=real(Ir);
B=imag(Ir);

printf("-Received current is Ir = %f /_%f mA\n",round(abs(Ir)*(10^3)*100)/100,fix(atan(B,A)*180*10/%pi)/10);

```

OUTPUT:

```
-Receiving end voltage Vr=0 because the receiving end has been short circuited
-Received current is Ir = 13.640000 /_-22.500000 mA

-->
```